

Unit III Graph Theory

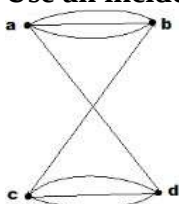
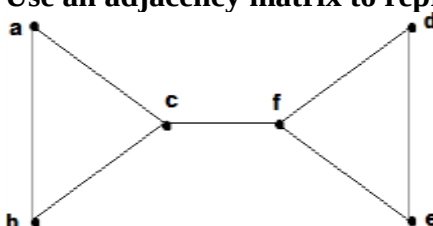
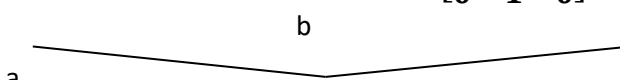
Course Objective – Cob3: To enable students understand basics of graph theory and algorithms.

Course Outcome – CO3: To demonstrate efficient use of graph theoretic concepts in solving connectivity and shortest path problems

Bloom's Taxonomy Level: K1, K2, K3, K4

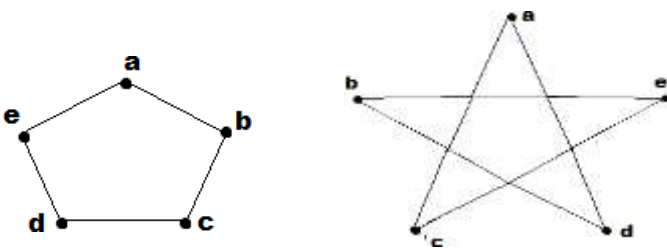
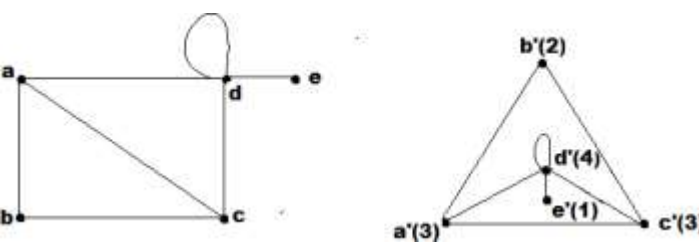
Part A

S.No.	Question	BTL*
1.	Define a graph. A Graph G is an ordered triplet $G = (V_G, E_G, \psi_G)$ where V –Set of all Vertices (Nodes/Points) E –Set of all Edges (Lines/Links) ψ –is an incidence map (function) that associates each edge $e \in E_G$ to an unordered pair of vertices $u, v \in V_G$.	K1
2.	Define simple graph, pseudo graph and multi-graph with example. Simple Graph: A graph is simple, if it has no loops (An edge with identical ends/vertex), parallel edges (two edges having same end vertices) and multiple edges (more than two edges having same end vertices). Pseudo Graph: A non-simple graph in which loops, parallel edges and multiple edges are permitted. Multi-Graph: A non simple graph in which only multiple edges are allowed	K1
3.	Define complete graph. Give an example. A simple graph in which each pair of distinct vertices is joined by an edge is called a complete graph and denoted by K_n , where n is number of vertices. Draw K_2, K_3, K_4, K_5 as an example.	
4.	Define bipartite graph. Give an example. A bipartite graph is one whose vertex set can be partitioned into two subsets X & Y , so that each edge has one end in X and other end in Y , such a partition (X, Y) is called a bipartition of the graph.	K1
5.	Define complete bipartite graph and draw graph of $K_{2,3}$ & $K_{3,3}$ A complete bipartite graph is a simple bipartite graph with bipartition (X, Y) in which each vertex of X is joined to each vertex of Y . If $ X = m, Y = n$, such a graph is denoted by $K_{m,n}$. Draw $K_{2,3}, K_{3,3}$ as an example.	K1
6.	Define regular graph & k-regular graph. Can a complete graph be a regular graph? Regular graph: A graph with all vertices have same degree then it is called regular graph k-regular graph: A graph G is k-regular if degree of each vertex should be k Can a complete graph be a regular graph? Yes, complete graph is a regular graph, since each vertex is of degree $n - 1$	K2

7.	Use an incidence matrix to represent the graph.  Incidence Matrix: <table><tr><th>Edges Vertices</th><th>e_1</th><th>e_2</th><th>e_3</th><th>e_4</th><th>e_5</th><th>e_6</th><th>e_7</th><th>e_8</th></tr><tr><td>a</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>b</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>c</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>d</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	Edges Vertices	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8	a	1	1	1	0	1	0	0	0	b	1	1	1	1	0	0	0	0	c	0	0	0	1	0	1	1	1	d	0	0	0	0	1	1	1	1	K2				
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8.	Use an adjacency matrix to represent the graph.  Adjacency Matrix: <table><tr><th>Vertices</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>a</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>b</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>c</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>d</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>e</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>f</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	Vertices	a	b	c	d	e	f	a	0	1	1	0	0	0	b	1	0	1	0	0	0	c	1	1	0	0	0	1	d	0	0	0	0	1	1	e	0	0	0	1	0	1	f	0	0	1	1	1	0	K2
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9.	Draw the graph with the following adjacency matrix $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ 																																																		
10.	State handshaking theorem. Let $G = (V, E)$ be an undirected graph with e – number of edges. Then $2e = \sum_{v \in V} \deg(v)$	K1																																																	
11.	How many edges are there in a graph with 10 vertices each of degree 5? We need to find $e = ?$ (No. of edges) Let $G = (V, E)$ with $V = \{v_1, v_2, v_3, \dots, v_{10}\}$ with $\deg(v_i) = 5, 1 \leq i \leq 10$ By H.S.T, $2e = \sum_{v \in V} \deg(v)$$\Rightarrow 2e = \sum_{i=1}^{10} \deg(v_i)$$\Rightarrow 2e = \deg(v_1) + \deg(v_2) + \dots + \deg(v_{10})$$\Rightarrow 2e = 5 + 5 + \dots + 5 \text{ (10 times)}$	K1																																																	

	$\Rightarrow 2e = 10 \times 5$ $\Rightarrow 2e = 50$ $\Rightarrow e = 25$	
12.	Define isomorphism between graphs. Two Graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be isomorphism, denoted by $G_1 \approx G_2$, if $\exists$ a 1 – 1 and onto function (1 – 1 correspondance) between G_1 & G_2 .	K1
13.	Define Euler Graph and Hamilton Graph. Give an example of a Euler graph and Hamilton Graph. Euler Graph: A connected graph with an Euler circuit is called an Euler graph. Hamilton Graph: A connected graph with a Hamilton circuit is called an Hamilton graph.	K1
14.	State the necessary and sufficient conditions for the existence of a Eulerian path in a connected graph. A connected graph is Eulerian if and only if Every vertex of G is of even degree	K1

Part B

S.No.	Question	BTL *
1.	Prove that the number of odd degree vertices in any graph is even.	K4
2.	State and prove handshaking theorem. Also prove that $\frac{n(n-1)}{2}$ is the maximum number of edges in a simple graph with n vertices.	K4
3.	Prove that a simple graph with n vertices and k components cannot have more than $\frac{[n-k][n-k+1]}{2}$ edges.	K4
4.	Define isomorphism between two graphs. Find whether the following graphs are isomorphic or not. 	K4
5.	Show that the following graphs are isomorphic. 	K4
6.	Examine whether the following pair of graphs are isomorphic or not. Justify your answer.	K4

7.	Prove that a connected graph G is Euler graph if and only if every vertex of G is of even degree.	K4
8.	Define Euler Graph and Hamilton Graph. Give an example of a graph (1) which is Eulerian but not Hamiltonian (2) Hamiltonian but not Eulerian (3) both Eulerian and Hamiltonian (4) neither Eulerian nor Hamiltonian.	K4

